

Trageted Node to be delete



```
def delete_node(self,node = None):  
    find = self.search_node(node)  
  
    if find is None:  
        return False  
    else:
```

For 0 child case :

_____ : 0 child case : _____

parent_find = find.parent

if find.left is None and find.right is None:

if find is self.root :

self.root = None

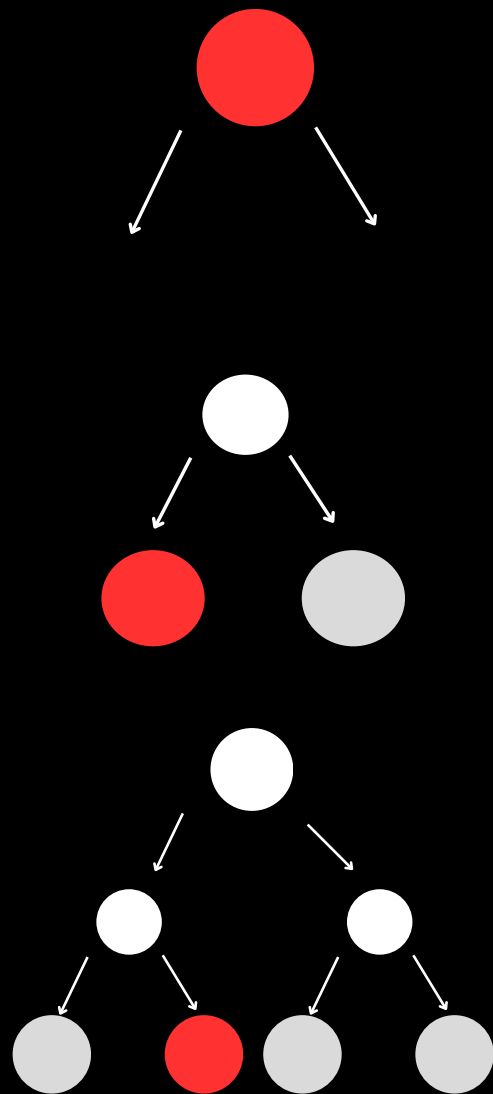
self.root.parent = None

elif find is parent_find.left:

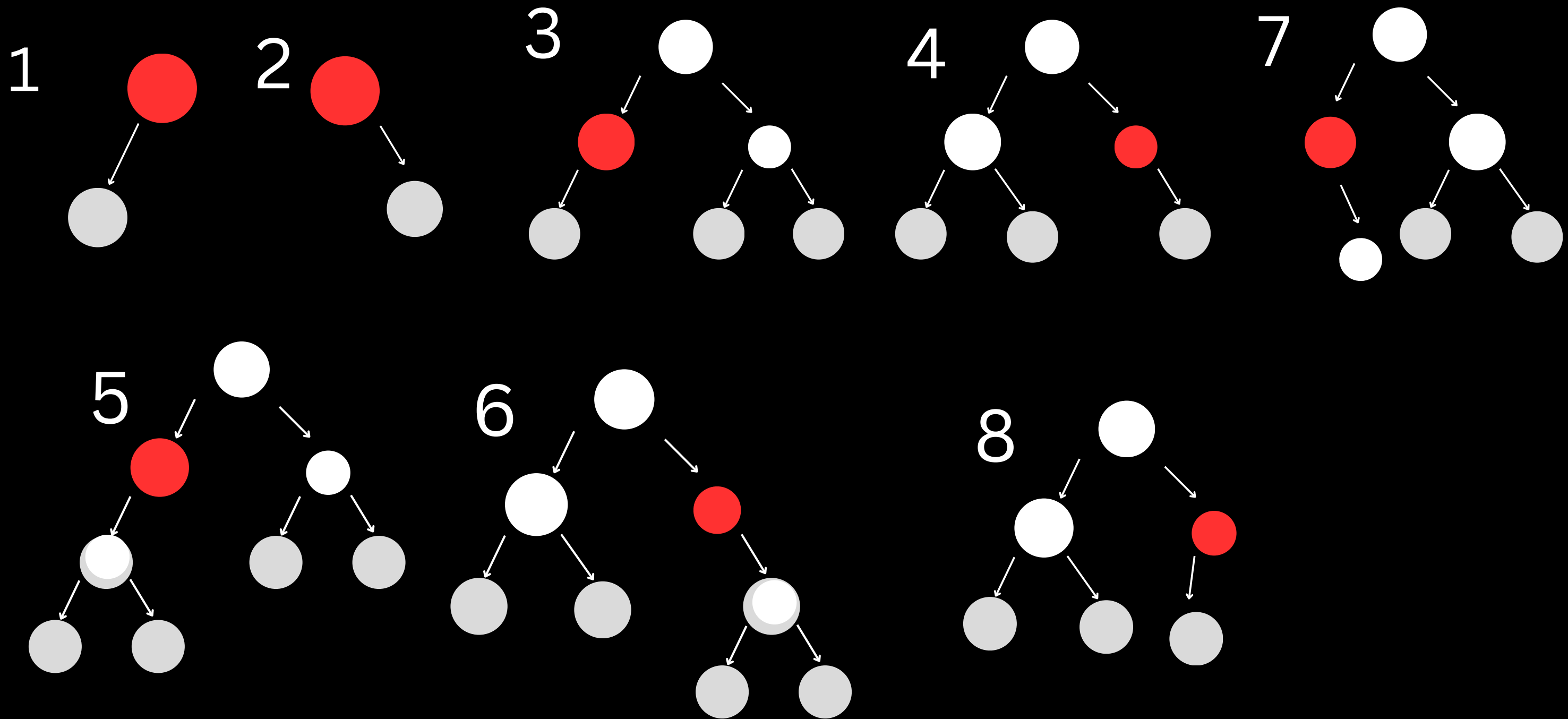
parent_find.left = None

else: #find is parent_find.right:

parent_find.right = None



For 1 child case :



Removing node which have 1 right child

elif find.left is None:

child = find.right

if find is self.root :

self.root = child

self.root.right = None

child.parent = None

self.root.parent = None

elif find is parent_find.left:

parent_find.left = child

child.parent = parent_find

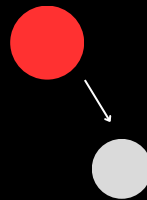
else: # find is parent_find.right:

parent_find.right = child

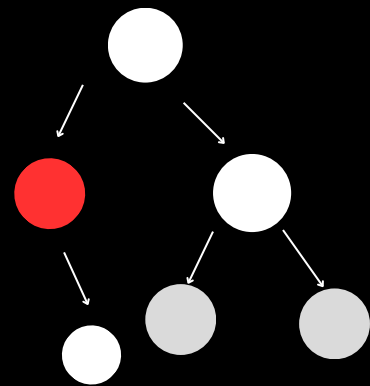
child.parent = parent_find

return True

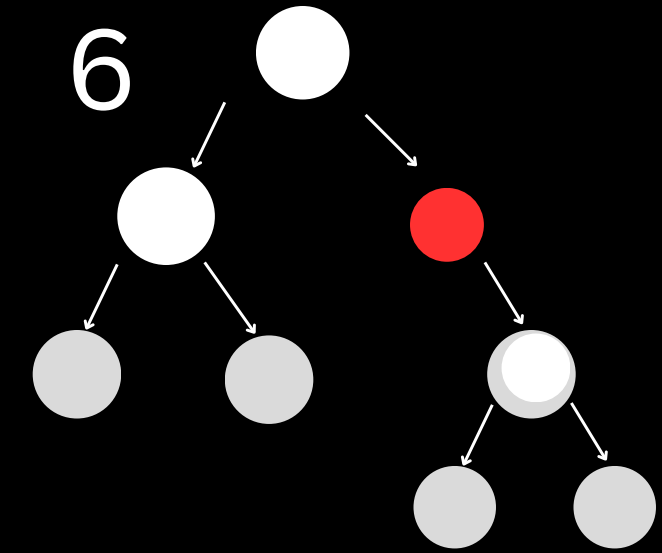
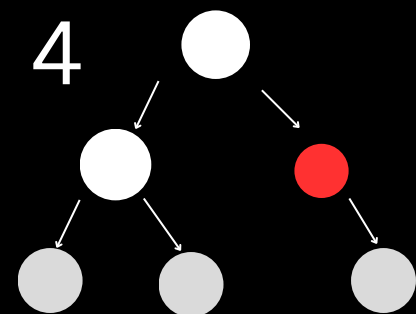
2



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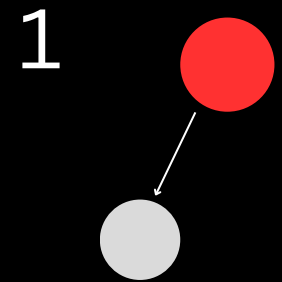


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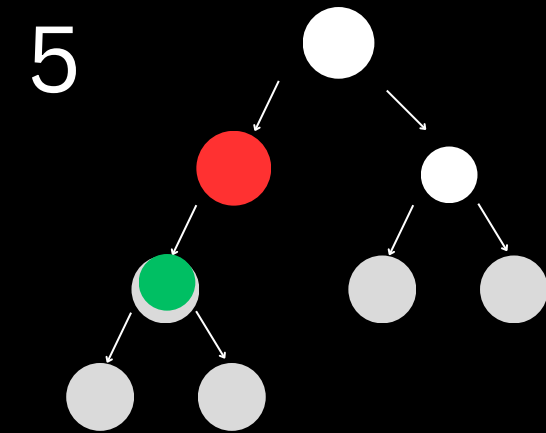
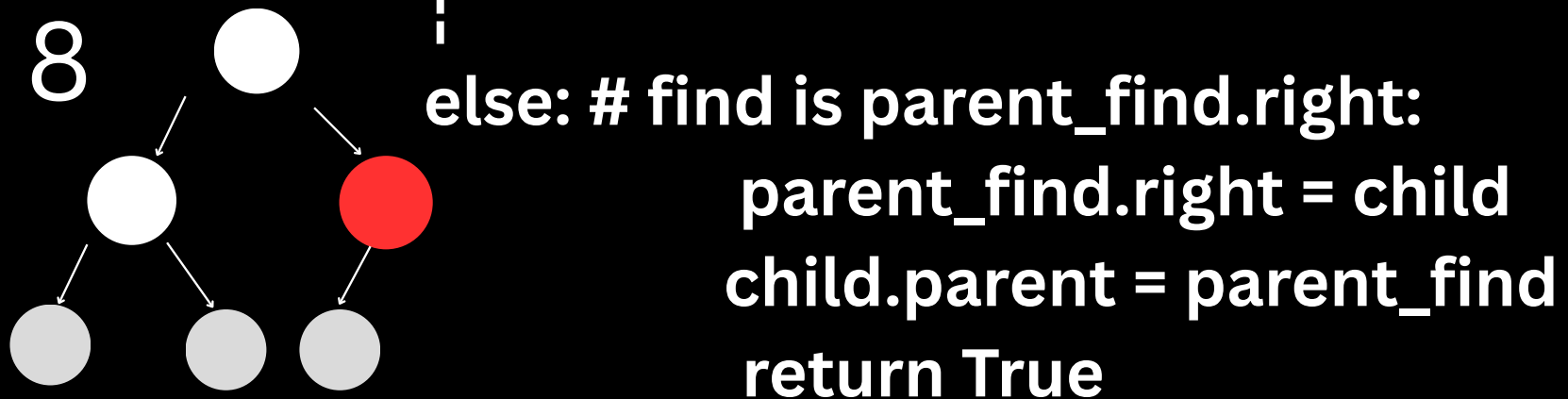
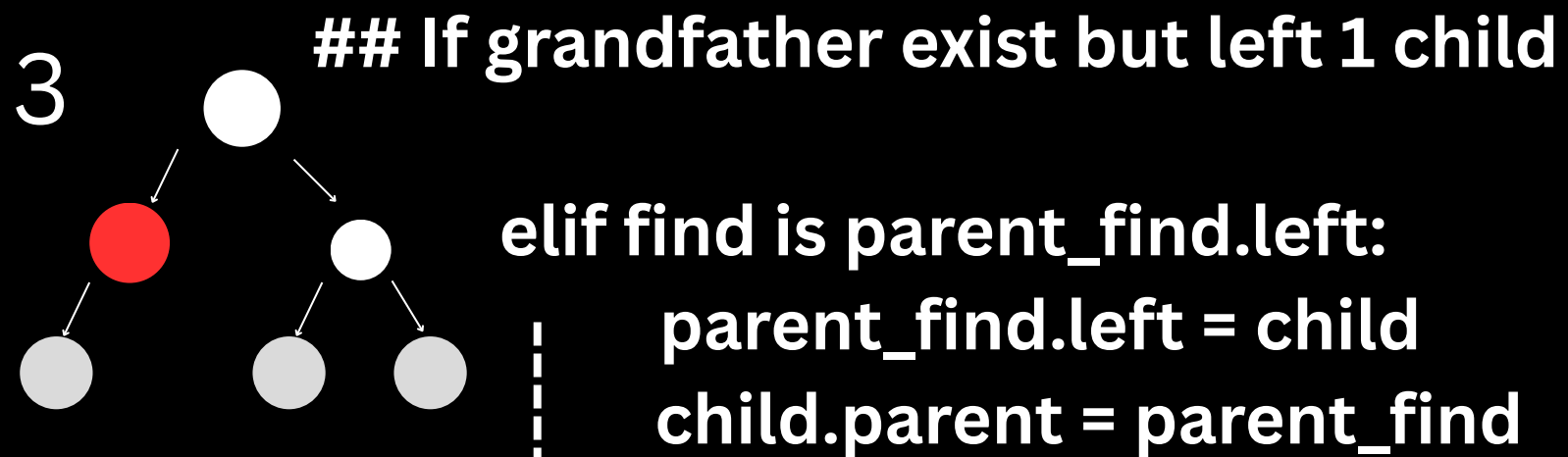


Removing node which have 1 left child

elif find.right is None:

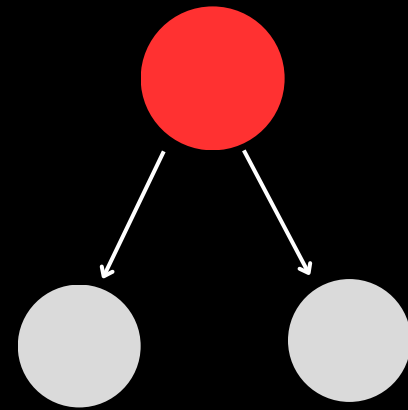


if find is self.root :
self.root = child
self.root.left = None
child.parent = None
self.root.parent = None

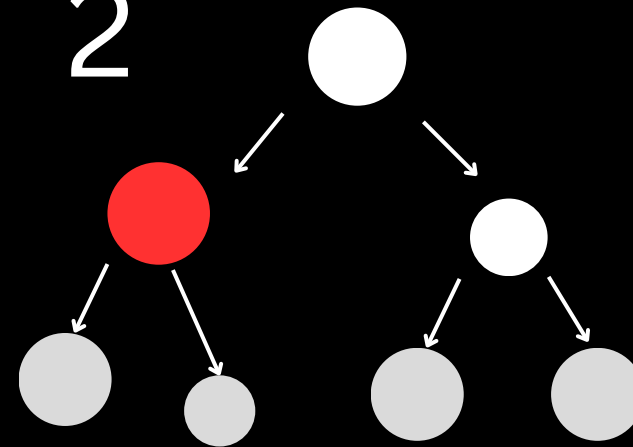


For 2 child case :

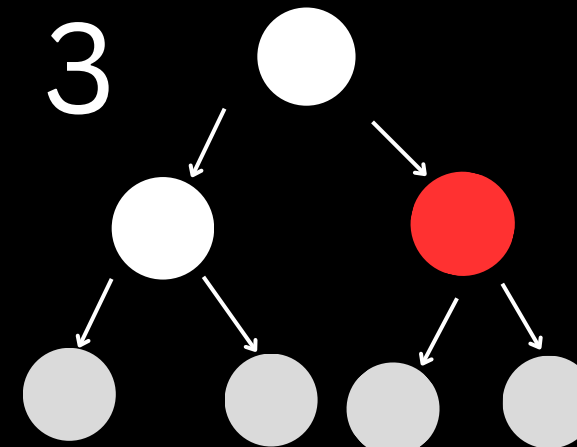
1



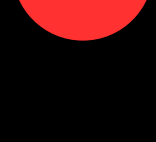
2



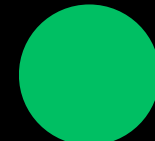
3



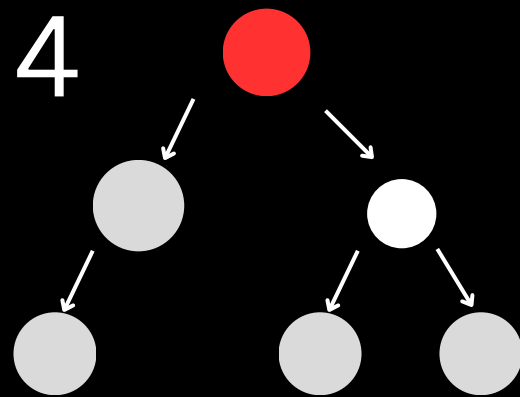
Trageted Node to be delete

None:  
Find  

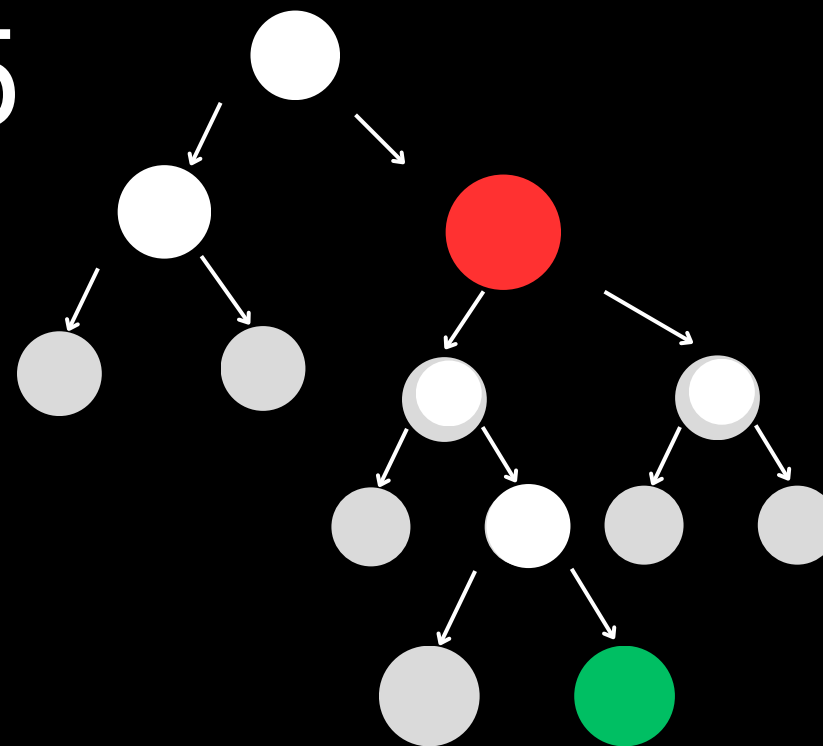
Trageted Node to be delete

Max  

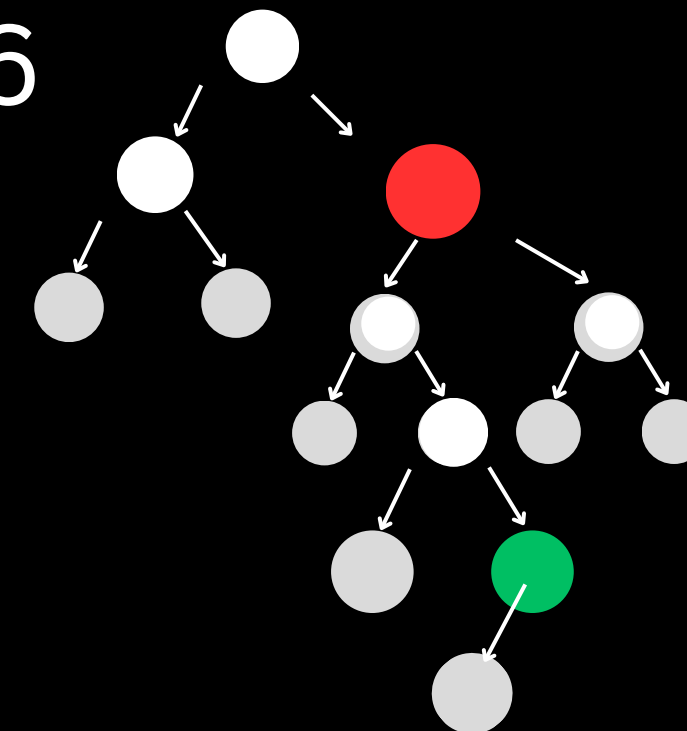
4



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_____ : 2 child case : _____

elif find.left is not None and find.right is not None:

Finding the maximum value from the left sub tree

max_value = self.max_value(find.left)

storing the parent of maximum value

max_vale_parent = max_value.parent

Copying the max value in targeted node

find = max_value

Storing the child of max_value if its left child exist

max_value_child = max_value.left

Deleting the node maximum value reference

if max_value is max_vale_parent.left:

max_vale_parent.left = max_value_child

else: # max_value is max_vale_parent.right

max_vale_parent.right = max_value_child

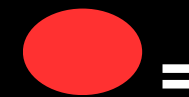
if max_value_child is not None:

max_value_child.parent = max_vale_parent

return True

else:

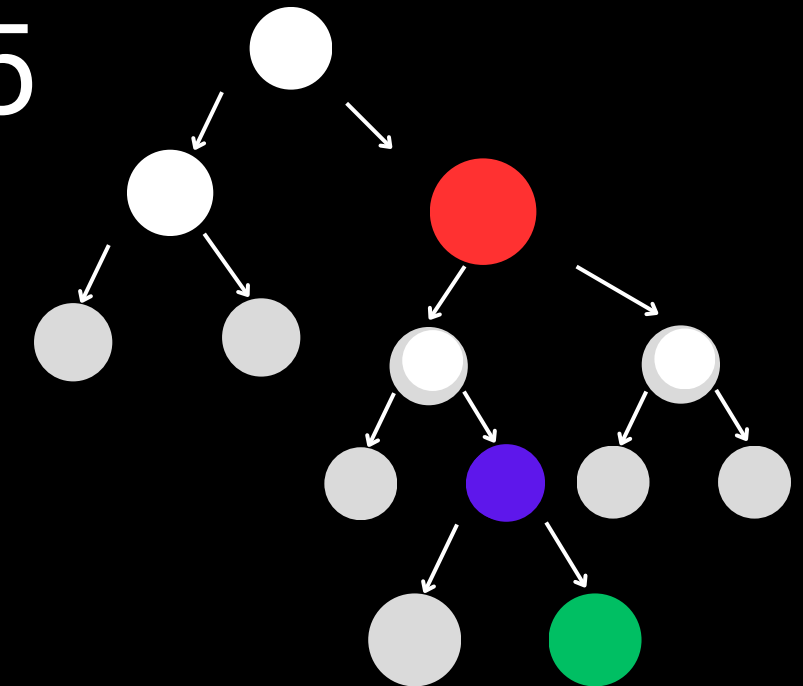
return False



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